5.17 Solution: (a) Similar to the discussion of Section 4.4, we can introduce image currents for the magnetic induction for $z > 0$,

$$\mathbf{B}_> = \frac{\mu_0}{4\pi}\int\left[\frac{\mathbf{J}(x',y',z')\times\mathbf{R}_1}{R_1^3} + \frac{\mathbf{J}'(x',y',-z')\times\mathbf{R}_2}{R_2^3}\right]d^3x,$$

and for $z < 0$

$$\mathbf{B}_< = \frac{\mu_r\mu_0}{4\pi}\int\frac{\mathbf{J}''(x',y',z')\times\mathbf{R}_1}{R_1^3}d^3x,$$

with

$$\mathbf{R}_1 = (x-x', y-y', z-z'), \quad \mathbf{R}_2 = (x-x', y-y', z+z').$$

At the boundary $z = 0$, we must have

$$B_{>,z}(x,y,0) = B_{<,z}(x,y,0), \quad H_{>,x(y)}(x,y,0) = H_{<,x(y)}(x,y,0).$$

Explicitly calculate the cross products, we have

$$\begin{aligned}
\mathbf{J}(x',y',z')\times\mathbf{R}_1\big|_{z=0} &= (-J_yz' - J_z(y-y'), J_z(x-x') + J_xz', J_x(y-y') - J_y(x-x')),\\
\mathbf{J}'(x',y',z')\times\mathbf{R}_2\big|_{z=0} &= (J_y'z' - J_z'(y-y'), J_z'(x-x') - J_x'z', J_x'(y-y') - J_y'(x-x')),\\
\mathbf{J}''(x',y',z')\times\mathbf{R}_1\big|_{z=0} &= (-J_y''z' - J_z''(y-y'), J_z''(x-x') + J_x''z', J_x''(y-y') - J_y''(x-x')).
\end{aligned}$$

Put these into the boundary conditions, we will have

$$J_x - J_x' = J_x'', \quad J_y - J_y' = J_y'', \quad J_z + J_z' = J_z'', \quad J_x + J_x' = \mu_r J_x'', \quad J_y + J_y' = \mu_r J_y'',$$

which leads to

$$J_x' = \frac{\mu_r - 1}{\mu_r + 1}J_x, \quad J_y' = \frac{\mu_r - 1}{\mu_r + 1}J_y, \quad J_x'' = \frac{2}{\mu_r + 1}J_x, \quad J_y'' = \frac{2}{\mu_r + 1}J_y.$$

To determine the the z-components of the image currents, we can invoke the continuity condition,

$$\nabla\cdot\mathbf{J} = 0.$$

Notice that $\mathbf{J}'$ is located at $(x', y', -z')$, we then have

$$J_z' = -\frac{\mu_r - 1}{\mu_r + 1}J_z, \quad J_z'' = \frac{2}{\mu_r + 1}J_z.$$

(b) It is obvious from part (a) that the magnetic induction in the region with $z < 0$ is given by

$$\begin{aligned}
\mathbf{B}_< &= \frac{\mu_r\mu_0}{4\pi}\int\frac{\mathbf{J}''(x',y',z')\times\mathbf{R}_1}{R_1^3}d^3x\\
&= \frac{\mu_0}{4\pi}\int\frac{2\mu_r}{\mu_r + 1}\mathbf{J}(x',y',z')\times\frac{\mathbf{R}_1}{R_1^3}d^3x,
\end{aligned}$$

which is equivalent to the magnetic induction induced by a current distribution of

$$\frac{2\mu_r}{\mu_r + 1}\mathbf{J}$$

in a medium with unit relative permeability.